

ALU's output is the result of the performed operation. In many designs, the ALU also exchanges additional information with a status register, which relates to the result of current or previous operations. The pin diagram of the ALU is shown in Fig. 1 and its description in Table I. The ALU architecture is shown in Fig. 2 and its function tables are listed in Table II.

Software program

ModelSim is an easy-to-use yet versatile VHDL/SystemVerilog/Verilog/

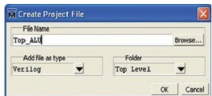


Fig. 5: Create Project File window

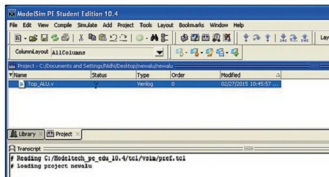


Fig. 6: Workspace window

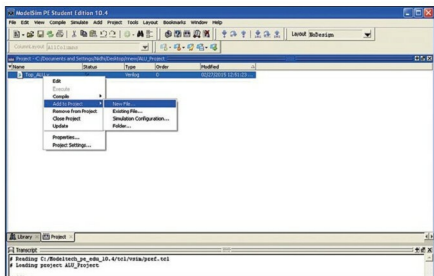


Fig. 7: Adding new files

SystemC simulator by Mentor Graphics. It supports behavioural, register-transfer-level and gate-level modelling.

First, install ModelSim on a Windows PC.

1. Start ModelSim from desktop; you will see ModelSim 10.4 dialogue window.
2. Create a project by clicking Jump-start on the Welcome screen.
3. A Create Project window pops up (Fig. 3). Select a suitable name for your project.



Fig. 8: Workspace section

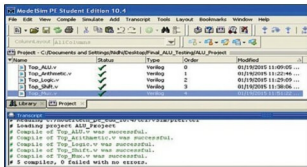


Fig. 9: Compilation window

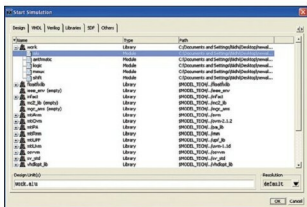


Fig. 10: Library tab

Set Project Location to C:/Documents and Settings/Nidhi/Desktop/Final_ALU_Testing (in our case) and leave the rest as default, followed by clicking OK.

3. An Add items to the Project window pops up (Fig. 4).
4. On this window, select Create New File option.
5. A Create Project File window pops up. Select an appropriate file name (say, `Top_ALU`) for the file you want to add; choose Verilog as Add file as type and Top Level as Folder (Fig. 5).
6. On the workspace section of the main window (Fig. 6), double-click on the file you have just created